

Codefest 1: Solutions

Hardware for Artificial Intelligence and Machine Learning (HW4AI)

ECE 410/510

Spring 2026

S/NY criteria

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CMAN — S requires: per-layer MAC formula shown; total MACs = 234,752 $\pm 5\%$; total params = 234,752 $\pm 5\%$; weight memory = 939,008 bytes $\pm 5\%$; activation memory = 4,712 bytes $\pm 5\%$ (or equivalent with stated assumption); arithmetic intensity computed from stated values; compute-bound/memory-bound conclusion consistent with calculated value vs. stated ridge point.

CLLM — S requires: GitHub repo is public and accessible; resnet18_profile.txt present and non-empty; top-5 MAC layers identified with names and values from torchinfo output; arithmetic intensity shown with formula for top layer; the first 3 Heilmeier questions present with ≥ 2 sentences each; block diagram present as an image file.

NY if: any required file missing; CMAN steps not shown; fewer than 3 Heilmeier answers; GitHub repo private or missing.